FreqDuet: Online Frequency-to-Authority Allocation for Hierarchical Bus Control

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Abstract

Terminal dispatch and intermediate-stop holding act at different scales, but a learned hierarchy must still decide which demand information each authority can use. FreqDuet updates a harmonic demand model only after automatic passenger counting bins close. It routes low-frequency level, slope, and forecast information to an executable terminal planner, and station-local innovations with compact passenger and vehicle context to discrete holding. In an independent 40-episode confirmation with 24 paired rollouts, the complete controller changed headway coefficient of variation by -0.022 [-0.038, -0.008] relative to a same-protocol reference. The bootstrap interval excluded zero, whereas the Holm-adjusted training-seed sign-flip result was $p = 0.125$. Restricted passenger journey changed by -0.263 [-0.837, +0.175] min and met the registered no-harm condition. In a separate 200-episode test with 64 pairs, journey improved by -1.242 [-2.204, -0.536] min, but the headway effect weakened to -0.009 [-0.028, +0.006] and failed the registered long-training gate. A preregistered development screen of three lower-actor freeze schedules produced no passing repair. Against fixed headway, FreqDuet was more regular but increased passenger journey by +2.470 [+1.874, +3.188] min. The evidence supports an online frequency-to-authority architecture and a bounded short-training result. It does not establish long-run regularity, passenger superiority over fixed headway, or an isolated causal effect of frequency allocation.

Keywords: bus holding, hierarchical reinforcement learning, demand decomposition, causal observation, headway control, reproducibility

1. Introduction

Frequent bus services are vulnerable to self-reinforcing irregularity. A bus that is delayed encounters more passengers, incurs longer dwell times, and can fall further behind, while the following bus experiences the opposite feedback. Holding and dispatch control can interrupt this process, but they exchange headway regularity for in-vehicle delay, operating time, and fleet availability (Daganzo, 2009; Xuan et al., 2011). Comparative studies consequently quantify trade-offs among holding strategies across regularity and holding burden (Berrebi et al., 2018), and reviews continue to identify passenger-oriented, real-time, and coordinated control as central research needs (Ibarra-Rojas et al., 2015; Gkiotsalitis and Cats, 2021). This trade-off makes a strong fixed-headway policy an essential comparator rather than a nominal baseline.

Reinforcement learning (RL) can optimize nonlinear bus operations without solving a new mathematical program at every control event. Existing work has learned fleet-wide holding policies, represented asynchronous bus arrivals, modeled uncertainty with distributional objectives, and combined holding with other control actions (Wang and Sun, 2020, 2021, 2023; Rodriguez et al., 2023). Hierarchical multi-agent control has also separated strategic action selection from detailed holding or speed decisions (Yu et al., 2024). These studies establish that learning-based control can coordinate event-driven transit operations, while a recent systematic evaluation shows that observation design, discrete action sets, and evaluation configuration can be as consequential as the selected RL architecture (Xu et al., 2025). A remaining design question is how a learned hierarchy should allocate exogenous demand information between a slower planner and a faster station-level controller.

Temporal abstraction supplies one part of the answer. Semi-Markov options and goal-conditioned hierarchical RL separate decisions by duration or authority (Sutton et al., 1999; Nachum et al., 2018), and slow representations can improve high-level subgoal selection (Li et al., 2021). Multi-frequency RL instead studies actions that persist or update at different rates (Lee et al., 2020; Metelli et al., 2020; Holt et al., 2025). Adjacent policy-learning work uses Fourier or wavelet representations to expose long trends and local detail (Huang et al., 2025; Duan et al., 2025). These approaches motivate multi-scale decision making, but temporal hierarchy, action frequency, and signal decomposition are not interchangeable. In particular, they do not by them-

selves specify which component of an external demand stream is available to each control authority without future-information leakage.

This study examines that interface in an asynchronous bus-control system. We propose FreqDuet, an online frequency-to-authority architecture. It updates a harmonic demand model from completed automatic passenger counting (APC) bins. Low-frequency level, slope, and forecast information enters an executable terminal headway planner. Station-local innovations and compact APC and automatic vehicle location (AVL) context enter a discrete holding policy. The two authorities remain coupled through the executable target headway, while planned and actual terminal releases remain distinct under a fixed physical fleet. We make three contributions. First, we formulate frequency-to-authority allocation as an explicit information interface for asynchronous hierarchical control. Second, we implement that interface with online observation timing, executable timetable semantics, and coherent sampled holding actions. Third, we evaluate a frozen controller in independent short- and long-training stages, retain the failed robustness result, and report a preregistered timescale-repair screen in which no candidate passed.

The empirical claim is deliberately narrower than the architectural proposal. The confirmatory contrast changes compact APC/AVL context and a two-sided regularity reward while retaining the harmonic frequency pathway in both arms. It therefore tests the complete selected controller, not the isolated causal effect of frequency separation. The study asks whether that controller has a reproducible operating effect and where the effect stops, rather than treating all favorable metrics or development variants as confirmation. In particular, the V33 repair screen remains negative evidence and does not supersede the failed long-training gate.

2. Related Work

2.1. Timetable, dispatch, and holding control

Classical bus-control research treats instability through schedules, dispatch rules, holding, speed control, or combinations of these interventions. Dynamic headway control can reduce bunching with less schedule slack than conventional schedule adherence (Daganzo, 2009), whereas virtual-schedule control can jointly target schedule reliability and regular headways (Xuan et al., 2011). Prediction-based holding can improve the compromise between

regularity and holding time, but its performance depends on prediction quality and operating configuration (Berrebi et al., 2018). Broader reviews organize these methods by planning horizon, control action, objective, and real-time feasibility (Ibarra-Rojas et al., 2015; Gkiotsalitis and Cats, 2021). The common operational lesson is that regularity is not free: holding can protect downstream headways while delaying onboard passengers and occupying vehicles.

FreqDuet retains this physical accounting. Its upper action changes an executable terminal headway plan, and its lower action delays a vehicle at an intermediate stop. These are different control locations, but both alter future departure gaps. The evaluation therefore reports passenger journey, realized holding, delayed fleet readiness, launch completion, and headway variation separately. This differs from interpreting a regularity-weighted scalar as a complete measure of passenger or operator performance.

2.2. Learning-based control for asynchronous bus operations

Deep RL has expanded bus holding from local rules to coordinated policies. Fleet-level multi-agent RL captures interactions that are difficult to encode in a fixed local law (Wang and Sun, 2020). Event-driven formulations address the fact that buses reach control stops asynchronously and that other agents may act between two decisions of a focal bus (Wang and Sun, 2021). Distributional and meta-learning extensions target demand surges, traffic perturbations, and service interruptions (Wang and Sun, 2023), while cooperative formulations combine holding with stop skipping and explicitly model nearby agents (Rodriguez et al., 2023). Hierarchical multi-agent RL has separated a high-level choice of intervention from a lower-level action magnitude (Yu et al., 2024).

This literature also cautions against equating a larger model with a stronger controller. A systematic bus-bunching evaluation found that compact spacing states can outperform more elaborate states and that discrete holding can match continuous actions while being easier to implement (Xu et al., 2025). FreqDuet accordingly uses event-specific states and a small discrete holding alphabet. Its distinction is not a new generic multi-agent critic. Instead, it defines an information boundary between a dispatch-scale planner and an arrival-scale holding policy, then evaluates the resulting controller under shared scenario tapes and fixed fleet semantics.

2.3. Hierarchical and frequency-aware decision making

Hierarchical RL traditionally separates behavior through temporally extended options, learned goals, or subgoal representations (Sutton et al., 1999; Nachum et al., 2018). Slow-feature objectives make this temporal intuition explicit by learning high-level representations that change slowly enough to support abstract exploration (Li et al., 2021). These methods concern the organization of decisions or endogenous state representations; they do not necessarily split an external time series into signals with different control ownership.

A second line of work models control rates directly. Action-persistence methods adapt how long an action is repeated (Metelli et al., 2020), multi-frequency RL handles action variables that update at different periods (Lee et al., 2020), and bi-level high-frequency control learns slow and fast policies jointly (Holt et al., 2025). Their central object is the rate at which actions are selected. In FreqDuet, asynchronous action rates are already fixed by transit events; the additional question is which part of observed demand is available to each policy.

A third line applies spectral representations to policy inputs, context, or action sequences. Learnable wavelet policies decompose long-horizon sequences into coarse and fine components (Huang et al., 2025), and low-frequency truncation has been used to summarize long context for multi-agent RL (Duan et al., 2025). FreqDuet adopts the multiscale motivation but uses an online harmonic estimator with a historical prior rather than a full-trajectory transform. The estimator is causal only in the temporal sense: it uses no future observations. Low-frequency demand summaries inform upper planning; station-local pre-update innovations inform lower holding. This allocation is a control contract, not merely a richer shared representation. The current experiment, however, compares complete policies and does not isolate this contract from the simultaneous APC/AVL and reward changes.

2.4. Passenger data and realism

Automated fare collection (AFC), APC, and AVL systems provide complementary views of passenger demand and vehicle motion. Smart-card records support strategic, tactical, and operational analysis, but their fields and sampling processes vary by agency (Pelletier et al., 2011). FreqDuet uses local historical OD intensities to initialize its demand prior. Public MTA AFC and Halifax APC subsets are used only to compare normalized

demand shapes; they do not share the evaluated network, service day, or measurement process. The external-data analysis is therefore a realism audit, not policy validation on those agencies.

3. Methods

3.1. Study question and control setting

FreqDuet allocates two causally available components of an exogenous demand stream to two authorities in an asynchronous hierarchical controller. Here, `causal` refers only to online information timing: a decision cannot use an observation that has not yet occurred. It does not denote identification of a causal treatment effect. The upper policy changes an executable target-headway plan at dispatch events, whereas the lower policy chooses holding at station-arrival events. One upper decision can therefore span many lower decisions. The current paper controller is the compact APC/AVL, weight-two configuration that passed the preregistered V8 gate. Its exact identifier and inheritance chain are reported in Supplementary Section S2.

3.2. Simulation environment and common random numbers

The simulator advances in 1-s steps on one bidirectional corridor with 22 physical stops (two terminals and 20 intermediate stops) and 42 directed inter-stop links. Each direction is 10.5 km long, comprising 21 links of 500 m. The input timetable contains 262 trips, split equally between directions. Service operates from 06:00 to 19:00 with a 4-hour clearance period, all scheduled trips, a fixed pool of 12 physical vehicles, and a capacity of 50 passengers per vehicle.

Passenger demand comes from a 20-origin by 14-hour by 20-destination historical OD-intensity table. Every 20 s, the simulator draws an independent Poisson count for each active OD cell and assigns each generated passenger a uniform arrival time within that window. A passenger remains latent until its assigned arrival time has elapsed. Each service-hour intensity is multiplied by a `Normal(1, 0.15)` draw clipped to `[0.3, 2.0]`; the historical peak profile is also shifted by -1, 0, or +1 hour with probabilities 0.2, 0.6, and 0.2. Segment speed limits update every 300 s by adding Gaussian variation with standard deviation 1.5 to the corresponding hourly route-history value, clipping the draw to `[2, 15]`, and applying the segment maximum. An inherited corridor calibration multiplies reverse-direction OD intensities at X13–X15 by 0.4 for every policy.

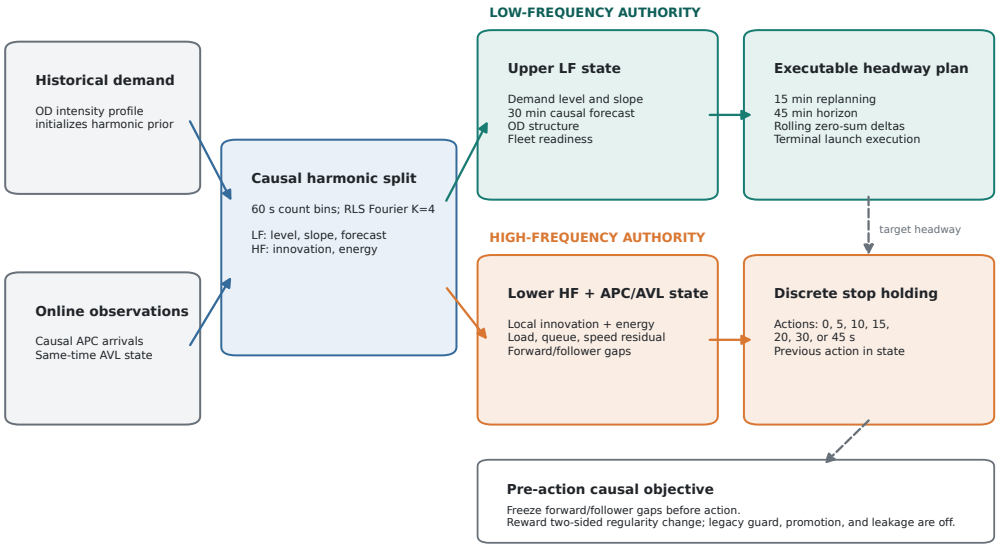


Figure 1: Causal frequency-to-authority architecture of the current controller. Historical OD intensities initialize a recursive harmonic demand prior, while online APC arrivals update the filter causally in 60-s bins. Low-frequency level, slope, and forecast features enter the upper policy, which replans an executable terminal headway curve every 15 min over a 45-min horizon under a rolling zero-sum headway budget. Station-local high-frequency innovations and compact same-time APC/AVL context enter the lower policy, which selects from seven discrete holding actions between 0 and 45 s. The two-sided regularity reward uses forward and follower departure gaps frozen before the action. In the current paper configuration, the legacy holding guard, promotion, and leakage penalty are disabled.

All exogenous draws are supplied by a policy-independent scenario tape keyed by evaluation seed and process identity. Passenger counts and arrival times, hourly demand multipliers, peak shifts, and fixed-clock route-speed streams therefore remain aligned across paired policies even when their action and learning call sequences differ. V8 and V9 use identical tapes within each policy pair. The public AFC/APC data are used only for the separate realism audit and do not calibrate the evaluated policy.

3.3. Causal harmonic demand decomposition

Observed APC arrivals are accumulated in 60-s bins. For bin `k`, the harmonic basis contains an intercept, a within-day linear trend, and `K=4` sine/cosine pairs over a 14-h period. Historical OD intensities fit a ridge-

regularized prior for `log(1 + arrival rate)` with ridge 0.01 and initial covariance 0.01. Recursive least squares with forgetting factor 0.9995 then updates that prior only after the current observation bin closes. With basis ϕ_k, coefficients θ_k, and pre-update prediction $\widehat{\lambda}_{k|k-1}$, the high-frequency innovation is

$$r_k = y_k - \widehat{\lambda}_{k|k-1}.$$

The low-frequency state is the updated nonnegative harmonic rate, its slope, and its 30-min forecast. The residual and its exponentially smoothed energy form the high-frequency state. Decisions at the start of a bin cannot observe arrivals later in that bin. This ordering, rather than a full-day transform, is the operational no-leakage guarantee.

3.4. Frequency-to-authority allocation

The upper policy receives global low-frequency level, slope and forecast, together with a scalar high-frequency energy summary and low-frequency OD structure summaries. The lower policy receives station-direction residual, residual change, local and global residual energy, the previous holding action, and compact same-time APC/AVL context: load, capacity, queue, speed residual, shock age, schedule slack, a normalized two-sided holding target, and a target-valid indicator. The scalar energy summary alerts the upper layer to volatility without giving it the local residual that drives holding.

The current controller does not use the historical promotion, leakage-penalty, or legacy causal holding-guard branches. Those mechanisms were development variants and are not part of the evaluated V6 policy. Consequently, the paper claim is the behavior of the complete current controller, not an isolated effect of promotion, leakage regularization, or guard removal.

3.5. Executable upper timetable

The upper ensemble actor produces a headway adjustment bounded to [-60, 60] s. Every 15 min, the timetable planner maps that action to an exact terminal headway curve over a 45-min horizon. The rolling zero-sum projection conserves the cumulative headway budget over each closed replanning window, preventing hidden phase drift. Planned launch times are executable: an actual departure occurs no earlier than both vehicle readiness and the scheduled launch time. Planned and actual terminal times are logged separately, and terminal shifts are bounded to [-45, 45] s.

3.6. Discrete lower holding and regularity reward

At each eligible station arrival, the lower categorical ensemble policy chooses holding from `{0, 5, 10, 15, 20, 30, 45}` s. Training samples from the policy; frozen evaluation uses its deterministic output. The chosen action is sent to the environment without a post-policy projection. The lower state includes the analytic balancing target derived from a matched predecessor departure and a same-time AVL estimate of the following vehicle.

Let `g_f` be the pre-action forward departure gap, `g_b` the same-time follower gap, `h` the executable target headway, and `a` the sampled hold. A hold predicts gaps `g_f + a` and `max(g_b - a, 0)`. For tolerance `tau=0.02`, define

$$q(g,h) = \left[\max\left(\frac{|g-h|}{h} - \tau, 0\right)\right]^2,$$
$$L(g_f, g_b, h) = \frac{1}{2}[q(g_f,h) + q(g_b,h)].$$

The lower reward receives $2\,\mathrm{clip}(L_{\mathrm{before}} - L_{\mathrm{after}}, -0.25, 0.25)$. All gaps are frozen before the action. Missing predecessor or follower evidence adds zero regularity reward and is logged rather than imputed from future vehicle states.

3.7. Learning architecture

Both levels use off-policy soft actor-critic variants (Haarnoja et al., 2018). The upper policy uses a pessimistic 10-critic ensemble, discount 0.95, a 64-unit hidden representation, batch size 64, and 10 updates per episode. The lower policy uses a pessimistic constrained 10-critic ensemble, discount 0.99, a 64-unit hidden representation, batch size 512, and 30 updates per episode. Both learning rates are `0.0003`; the lower dual learning rate is `0.0001`. The lower policy's learned constraint cost is optimized through a Lagrange multiplier, following the constrained-RL formulation (Miryoosefi et al., 2019). The upper policy begins after a 30-episode lower warm-up. V8 trains for 40 episodes and evaluates checkpoint 39; V9 trains without policy or critic freezing for 200 episodes and evaluates checkpoint 199. Both upper and lower actors are deterministic during frozen evaluation, and the evaluator rejects any change in deployment state across evaluation seeds.

V33 is a subsequent, preregistered development screen of lower-policy update duration. Each candidate trains for 200 episodes, but lower-actor

updates stop after episode 40, 80, or 100. The lower critics and both upper components keep training. V33 uses four new training seeds crossed with four new evaluation seeds. It is not an independent confirmation stage; its confirmation roster was reserved for use only if exactly one candidate passed the development gate.

3.8. External comparators

Three non-learned comparators use the same V6 environment, fixed 12-vehicle pool, timetable, evaluation seeds, scenario tapes, and exact terminal-release semantics as the learned controller. `fixed_headway` installs a 360-s terminal headway in both directions and commands zero intermediate holding. `rule_holding` uses the same 360-s terminal schedule and applies

$$a = \mathrm{clip}(360 - g_f, 0, 60),$$

where `g_f` is the observed forward headway in seconds. `rule_mpc` uses that same lower holding law and re-evaluates a 60-candidate time-of-day headway grid at dispatch events. Peak candidates are `{240, 300, 360, 420, 480}` s, off-peak candidates are `{360, 480, 600, 720}` s, and transition candidates are `{300, 360, 420}` s. Given a seeded episode-level demand proxy d, with $d \sim \mathrm{clip}(\mathcal{N}(1, 0.15), 0.3, 2.0)$, its slot-specific surrogate is

$$H_{\mathrm{ideal}} = \mathrm{clip}(360/d, 240, 600),$$
$$J(H) = \frac{H}{2} + 0.001 \max(0, H - H_{\mathrm{ideal}})^2 + 5 \max(0, 6000/H - 12)^2.$$

The minimizing candidate is converted to an exact launch sequence before simulation. This rule-MPC is a transparent low-fidelity comparator, not a claim to represent an optimally tuned or simulator-aware MPC. The fixed-headway policy is the strong external comparator in this study.

3.9. Outcomes

The primary passenger endpoint is restricted total journey time per generated passenger. Waiting is censored at the evaluation horizon for passengers not yet boarded; in-vehicle and total journey time are censored at that horizon for passengers not yet arrived. Secondary outcomes are restricted waiting time, restricted in-vehicle time, unserved-passenger rate, headway coefficient

of variation, realized vehicle and passenger holding, fleet-denial measures, terminal execution error, trip completion, and restricted service cost. Headway CV is the standard deviation divided by the mean over valid recorded headway events. With restricted waiting `W_R` in minutes, peak fleet `F`, fixed fleet budget `N`, headway CV `H`, unserved fraction `U`, and trip-completion fraction `Q`, the secondary scalar is

$$C_R = \frac{W_R}{10} + \frac{\max(F - N, 0)^2}{N} + H + 5U + 5(1 - Q).$$

The evaluated fixed-pool environment makes the overshoot term zero; this scalar does not directly charge holding or a delayed-readiness denial that is later retried. It is therefore a secondary summary and does not replace the passenger and physical outcomes.

3.10. Evaluation and inference

V8 is a preregistered independent 40-episode confirmation with six training seeds crossed with four untouched evaluation seeds (24 paired rollouts per policy). V9 is a separately preregistered 200-episode robustness test with eight new training seeds crossed with eight new evaluation seeds (64 paired rollouts per policy). The comparator is the no-guard Protocol V6 reference, whose exact identifier is reported in Supplementary Section S4. Both policies disable the legacy holding guard; the current policy additionally has compact APC/AVL context and the two-sided regularity objective, so their difference is a combined-policy contrast.

The V33 development screen uses four training seeds crossed with four evaluation seeds (16 paired rollouts per candidate). A candidate had to pass the unchanged V8 effect and mechanism gate, place the headway-CV interval below zero, satisfy the passenger-journey no-harm bound, and improve headway CV for at least three of four training seeds. No candidate passed, so the reserved confirmation roster was not opened.

Uncertainty uses a crossed bootstrap over training and evaluation seeds while sharing each evaluation-seed resample across paired policies. Two-sided sign-flip tests operate on training-seed mean differences, with Holm correction within each metric family. Lower values favor FreqDuet for every reported outcome. For each external comparator, its eight evaluation-seed realizations are crossed with the eight V9 learned training seeds, yielding 64 paired rows; the crossed analysis retains training seed as the independent policy-training unit. V8, V9, and V33 estimates are kept separate and are never pooled.

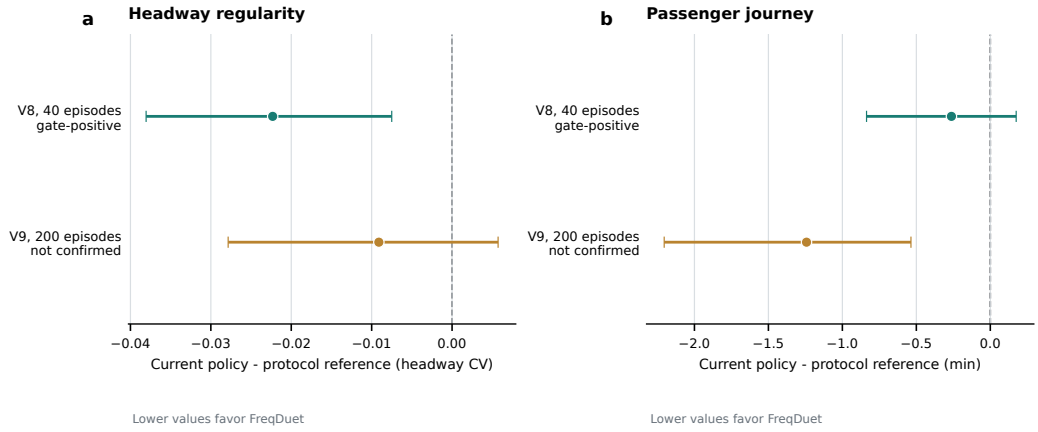


Figure 2: Independent confirmation and long-training robustness. Points show paired mean differences between the current policy and the Protocol V6 reference config named `F_freqduet_protocol_v6_noguard_hiro`; bars show 95% crossed-bootstrap confidence intervals over training and evaluation seeds. Both configurations disable the legacy causal holding guard. The current policy additionally uses compact APC/AVL context and the two-sided departure-regularity objective, so this is a combined-policy comparison rather than an isolated guard effect. Lower values favor the current policy. V8 contains 24 paired rollouts (six training seeds by four untouched evaluation seeds) and passed the registered effect/no-harm gate. Its Holm-adjusted training-seed sign-flip result was p=0.125, so the figure labels V8 as gate-positive rather than familywise significant. V9 contains 64 paired rollouts (eight by eight); its passenger-journey interval favored FreqDuet, but the headway-CV effect did not meet the registered magnitude and interval gate, so V9 is reported as not confirmed.

4. Results

4.1. Independent confirmation at 40 episodes

V8 passed the registered effect/no-harm gate for the complete current policy (Fig. 2; Table 1). Relative to the Protocol V6 reference, headway CV changed by -0.022 [-0.038, -0.008]. Its crossed-bootstrap interval excluded zero, whereas the Holm-adjusted training-seed sign-flip result was $p = 0.125$. Restricted passenger journey changed by -0.263 [-0.837, +0.175] min. The latter interval crossed zero but satisfied the preregistered journey no-harm margin. Thus V8 is gate-positive under its preregistered criteria; it is not a familywise-significant effect at 0.05 and does not establish a passenger-journey benefit.

Table 1. Current policy minus the Protocol V6 reference. Values are paired mean differences with crossed-bootstrap 95% confidence intervals. Lower is better. Holm-adjusted sign-flip p-values use training-seed mean differences.

Outcome	V8 delta [95% CI]	V8 Holm p	V9 delta [95% CI]	V9 Holm p
Restricted passenger journey (min)	-0.263 [-0.837, +0.175]	0.938	-1.242 [-2.204, -0.536]	0.047
Restricted passenger wait (min)	-0.185 [-0.470, +0.050]	0.375	-1.011 [-1.878, -0.397]	0.047
Restricted in-vehicle time (min)	-0.078 [-0.376, +0.147]	1.000	-0.232 [-0.355, -0.121]	0.047
Headway coefficient of variation	-0.022 [-0.038, -0.008]	0.125	-0.009 [-0.028, +0.006]	0.031
Unserved passengers (percentage points)	+0.02 [-0.00, +0.11]	1.000	+0.00 [+0.00, +0.02]	0.250
Realized holding (s/launched trip)	-1.7 [-41.8, +29.4]	1.000	-37.6 [-59.8, -16.2]	0.070
Trips denied at least once (percentage points)	+0.48 [-9.21, +8.81]	1.000	-1.65 [-3.96, -0.22]	0.117
Restricted service cost	-0.040 [-0.079, -0.007]	0.125	-0.110 [-0.207, -0.042]	0.047

4.2. Long-training robustness was not confirmed

At 200 episodes, the same policy improved restricted journey by -1.242 [-2.204, -0.536] min relative to the reference. The headway-CV difference was -0.009 [-0.028, +0.006]. Seven of eight training-seed CV differences were negative, but the interval included zero and the mean improvement did not reach the registered 0.02 threshold. V9 therefore returned `longtrain_not_confirmed`; the favorable journey result cannot be relabelled as confirmation of the registered regularity effect.

4.3. Lower-actor freezing did not repair long-training regularity

V33 tested three lower-actor freeze schedules on a preregistered 4-by-4 development roster. Freezing after episode 100, 80, or 40 changed headway CV by -0.007 [-0.049, +0.030], -0.004 [-0.031, +0.015], and -0.012 [-0.038, +0.008], respectively. Every interval crossed zero, and no mean reduction reached the registered 0.02 threshold. The episode-40 schedule improved restricted journey by -1.954 [-3.190, -1.077] min and had negative headway-CV differences for 4 of 4 training seeds. It still failed the registered magnitude and interval requirements. All three candidates therefore failed; the reserved confirmation roster was not used. The unfrozen controller passed the incremental gate on these development seeds, but it was a control rather than a V33 candidate and does not replace the failed independent V9 result. Full V33 outcomes are reported in Supplementary Section S5.

4.4. External baselines reveal a passenger-regularity trade-off

The source-identical V9 comparison (Fig. 3; Table 2) shows that FreqDuet had lower headway CV than fixed headway, -0.219 [-0.238, -0.198], and lower restricted service cost, -0.122 [-0.180, -0.050]. However, restricted journey was higher by +2.470 [+1.874, +3.188] min. FreqDuet also used +286.8 [+269.7, +302.0] more holding seconds per launched trip and had a +64.10 [+61.97, +66.03] percentage-point higher denied-trip rate. All trips were eventually launched and completed in the aggregated learned-policy results, so this denial measure captures delayed fleet readiness rather than permanent trip cancellation. Because the restricted service-cost scalar does not directly charge holding or retried readiness denials, its favorable difference cannot be interpreted as passenger-time or fleet-readiness superiority.

FreqDuet reduced restricted journey relative to rule holding by -3.224 [-4.727, -1.885] min and relative to rule MPC by -26.511 [-31.003, -21.628] min. The supported external conclusion is therefore narrower than universal

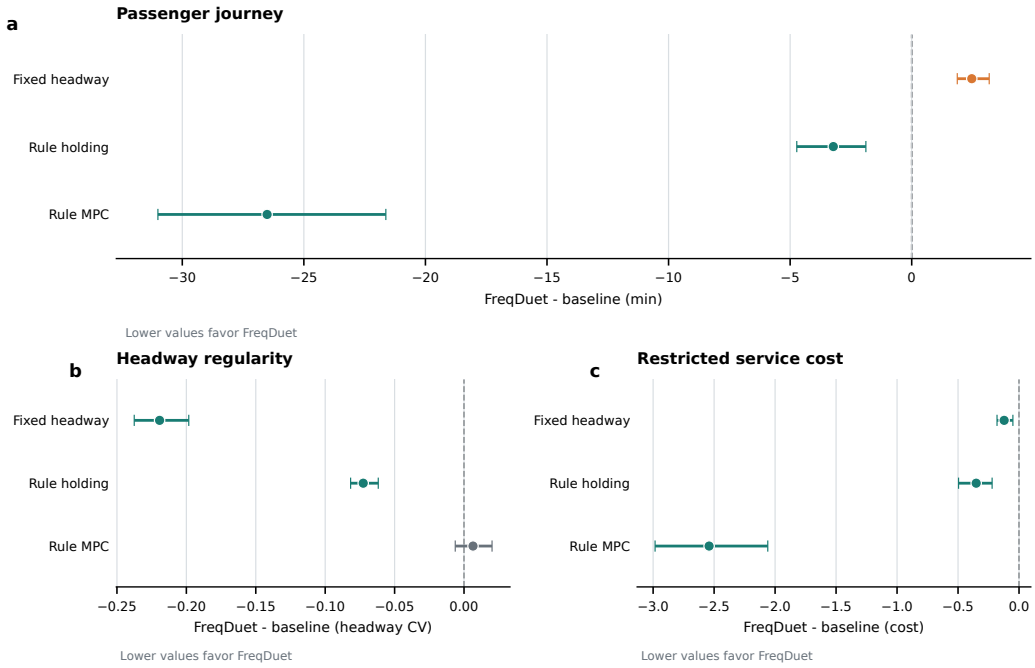


Figure 3: External baseline trade-off under the V9 source contract. Points show paired mean differences between FreqDuet and each external baseline; bars show 95% crossed-bootstrap confidence intervals over eight training and eight evaluation seeds (64 paired rollouts). Lower values favor FreqDuet. FreqDuet improved regularity and restricted service cost relative to fixed headway but increased passenger journey time. It reduced passenger journey time relative to rule holding and rule MPC. Exact two-sided sign-flip tests and Holm-adjusted values are provided in the source table and are not encoded as significance symbols in the figure.

superiority: FreqDuet reduced restricted journey relative to the two rules and produced more regular service than fixed headway, while fixed headway remained better for passenger journey and fleet-readiness burden.

Table 2. FreqDuet minus external baseline under V9. Values are paired mean differences with crossed-bootstrap 95% confidence intervals. Lower is better. The complete outcome and adjusted-test table is in the Supplementary Material.

Baseline	Journey min	Headway CV	Denied trips (pp)	Service cost
Fixed headway	+2.470 [+1.874, +3.188]	-0.219 [-0.238, -0.198]	+64.10 [+61.97, +66.03]	-0.122 [-0.180, -0.050]
Rule holding	-3.224 [-4.727, -1.885]	-0.073 [-0.082, -0.062]	-3.17 [-5.42, -1.90]	-0.351 [-0.496, -0.221]
Rule MPC	-26.511 [-31.003, -21.628]	+0.006 [-0.006, +0.020]	+61.52 [+59.09, +63.77]	-2.541 [-2.984, -2.060]

4.5. Physical execution audit

Relative to the Protocol V6 reference, V9 reduced realized holding by -37.6 [-59.8, -16.2] s per launched trip and the denied-trip rate by -1.65 [-3.96, -0.22] percentage points (Fig. 4). These are full-policy differences, not isolated effects of the regularity reward. Against fixed headway, however, the current policy used +286.8 [+269.7, +302.0] more holding seconds per launched trip and increased the denied-trip rate by +64.10 [+61.97, +66.03] percentage points. All trips were eventually launched and completed in the aggregated learned-policy results, so denial records delayed fleet readiness rather than permanent trip cancellation.

4.6. External data support demand-shape realism only

The FreqDuet OD input has a morning peak similar in timing to the bounded MTA AFC subset, while its normalized hourly profile differs from the Halifax APC subset (Fig. 5). The balanced audit contains 39 complete MTA station-complex days (936 rows) and seven complete Halifax routes across 37 route-days (979 rows). Because systems, dates, sampling units, and measurement processes are unmatched, these comparisons are descriptive checks of demand-shape plausibility. They are not same-day calibration, route-family policy tests, or field-effect estimates.

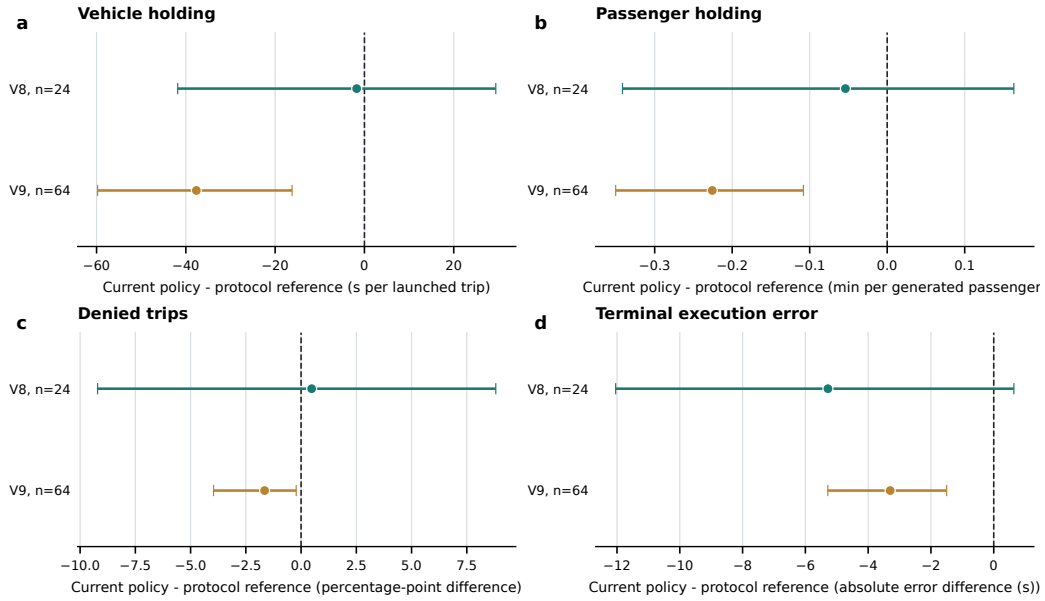


Figure 4: Paired physical outcomes of the current policy. Points show mean paired differences between the full current policy and the Protocol V6 reference configuration; bars show 95% crossed-bootstrap confidence intervals. Negative values favor the current policy. V8 contains 24 paired rollouts and V9 contains 64. The current policy differs from the reference by both compact APC/AVL context and the two-sided departure-regularity objective, so these panels describe the combined policy's physical behavior rather than an isolated legacy-guard effect.

5. Discussion

5.1. Principal findings

FreqDuet produced a positive result under its registered V8 gate, but the evidential scope is bounded. Under the independent V8 protocol, the frozen complete controller met the registered short-training regularity gate and the passenger-journey no-harm condition. The confidence interval for the paired headway-CV difference excluded zero, although the Holm-adjusted training-seed sign-flip result was $p = 0.125$. With six training seeds, the sign-flip randomization distribution is coarse. Under the separate V9 protocol, passenger journey improved relative to the same Protocol V6 reference (Holm-adjusted $p = 0.047$), but the registered long-training regularity magnitude was not confirmed. These stages answer different questions and are not pooled: V8 is gate-positive under its registered short-training criteria, whereas V9 did not recover its registered regularity magnitude under the longer learning horizon.

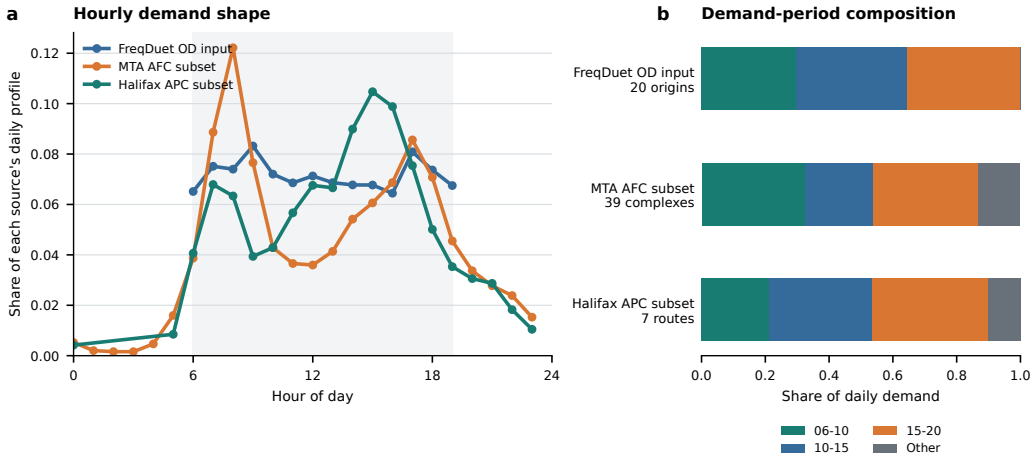


Figure 5: External passenger-count demand-shape audit. Panel a compares separately normalized hourly demand shapes from the FreqDuet OD input, a complete-day subset of the bounded public MTA AFC cache (936 source rows; 39 station-complex days), and a complete-route subset of the bounded public Halifax APC cache (979 source rows; 7 routes and 37 route-days). Panel b summarizes the corresponding demand-period shares; the FreqDuet input contains 20 origin series. The balanced-cache derivation excludes incomplete pagination fragments. This remains a descriptive audit across unmatched systems and dates, not a population estimate, same-day calibration, field-policy evaluation, or evidence of deployed control benefit.

The V9 headway sign-flip result was Holm-adjusted $p = 0.031$ even though the crossed-bootstrap interval included zero. These statistics use different sampling units: the sign-flip test reduces each trained policy to its mean over evaluation seeds, whereas the crossed bootstrap propagates variation from both training and evaluation seeds. The registered V9 gate required the latter effect-size interval and was not passed; the smaller sign-flip value is not used to relabel V9 as a confirmation.

V33 then tested a specific stability hypothesis: lower-actor updates after the short-training checkpoint might erode regularity. Freezing the lower actor after episode 40, 80, or 100 did not resolve the problem. All three schedules improved restricted journey relative to the reference, but no candidate passed the registered regularity gate. The episode-40 schedule had a negative headway-CV difference for every training seed, yet its mean reduction was only 0.012 and its confidence interval crossed zero. The unused confirmation roster therefore remained untouched. The unfrozen control passed on the V33 development roster, but that selected-data observation cannot replace

the failed independent V9 test.

This pattern matters for learning-based bus control. Prior work demonstrates that RL can coordinate asynchronous holding and adapt to operational uncertainty (Wang and Sun, 2020, 2021, 2023), but the present results show why training duration belongs in the evaluation contract. A policy can retain a favorable passenger outcome while the regularity effect used for selection weakens. V33 further shows that a plausible timescale repair can fail even when its passenger endpoint improves. Reporting only the selected checkpoint, the best development seed, or pooled training horizons would conceal that failure mode.

5.2. Passenger service versus regularity

The fixed-headway comparison exposes a second boundary. FreqDuet generated substantially more regular headways and a lower restricted service-cost scalar, but fixed headway delivered lower restricted passenger journey and imposed far less holding and delayed fleet readiness. This is consistent with the classical observation that holding trades regularity against onboard delay and vehicle productivity (Daganzo, 2009; Xuan et al., 2011; Berrebi et al., 2018). More regular service is therefore not sufficient to claim better passenger service.

The scalar service cost should also not be used to reverse this conclusion. In the evaluated fixed-pool environment its fleet-overshoot term is identically zero, and it does not directly price realized holding or a readiness denial that is later retried. The separate physical outcomes reveal burdens hidden by that summary. A passenger-centered deployment objective would need to price holding, readiness delay, and censored journey outcomes explicitly rather than relying on headway variation as a surrogate.

The two rule baselines bound narrower questions. The proportional controller is a transparent local holding law, and the rule-MPC uses a coarse time-of-day surrogate rather than a calibrated prediction model. Improvement over those rows does not establish superiority to the broader holding and model-predictive control literature. The fixed-headway row is the demanding external comparison for this experiment.

5.3. Interpretation of the frequency interface

The architecture gives low-frequency demand summaries to the upper planner and station-local innovations to the lower controller while preserving

causal observation timing. This is a concrete extension of temporal abstraction: the hierarchy separates not only when policies act, but also which component of an external process each policy can observe. The distinction complements work on temporally extended goals (Sutton et al., 1999; Nachum et al., 2018), multiple action frequencies (Lee et al., 2020; Metelli et al., 2020; Holt et al., 2025), and frequency-domain representations (Huang et al., 2025; Duan et al., 2025).

The experiments do not establish that this interface caused the V8 result. The current and reference controllers both retain the harmonic path, while the current controller also changes compact APC/AVL context and the two-sided regularity reward. Plausible explanations therefore include better local observability, denser action-aligned credit, their interaction with the frequency state, or ordinary training variation. The supported conclusion is about the complete controller. An isolated frequency claim requires a fresh, same-stage comparison against no-frequency, raw-history, all-frequency, and swapped-authority controls under the current physical semantics.

5.4. External realism and applicability

The MTA AFC and Halifax APC profiles show that the simulator demand is not a featureless stationary input and that its morning structure is plausible in at least one external comparison. They also show a mismatch with the separately normalized Halifax profile. That mismatch is informative because AFC entries, APC boardings, OD intensities, and vehicle locations are different measurement objects (Pelletier et al., 2011). The audit supports demand-shape plausibility only. It does not demonstrate transfer to another route or agency.

The implemented controller is most applicable to high-frequency services with terminal dispatch control, event-time AVL, station-level passenger observations, and a known physical fleet. Its causal filter can operate online because it updates only after a count bin closes. The current evidence does not establish performance with missing APC streams, multi-line vehicle sharing, driver noncompliance, or field dispatch interventions.

5.5. Limitations

Five limitations define the present claim. First, V9 did not confirm the registered 200-episode regularity effect, and no V33 freeze schedule repaired

it; long-horizon learning stability remains unresolved. Second, the confirmatory contrast is bundled and lacks current-stage NoFreq, RawHistory, AllFreq, and swapped-authority controls, so frequency allocation is not causally isolated. Third, the strong fixed-headway baseline is better on passenger journey and fleet-readiness burden, despite worse headway CV. Fourth, the policy is evaluated in one simulated network initialized from a local historical OD table. The public external data are unmatched descriptive profiles, not same-day AFC/APC/AVL calibration, route-family policy tests, or a field treatment effect. Fifth, the inherited simulator scales reverse-direction OD intensities at X13–X15 by 0.4. That calibration is common to all paired policies but was not re-estimated from the external data, further limiting transportability beyond the evaluated corridor.

5.6. Implications for further evaluation

The next decisive experiment is not another gate tuned on V8, V9, or V33. It is a new, frozen factorial confirmation that holds APC/AVL context, reward, physical fleet, and training protocol constant while varying only frequency allocation. Such a study should use unseen route and service-day families and retain fixed-headway passenger journey and fleet burden as co-primary constraints. Separately, replacing the current restricted scalar with an objective that prices holding and readiness delay would test whether the regularity gain can be converted into passenger benefit. Until those experiments are complete, FreqDuet should be read as a causally implemented and independently evaluated frequency-to-authority design with one registered gate-positive short-training signal, not as a universally superior bus controller.

6. Conclusions

FreqDuet implements an online frequency-to-authority interface for asynchronous bus control: historical and completed-bin demand information defines a smooth state for executable terminal headway planning, while station-local innovations and same-time APC/AVL context inform discrete intermediate-stop holding. The frozen controller met its independent short-training headway-regularity gate without violating the registered passenger-journey no-harm condition. In the separate long-training test, passenger journey improved relative to the Protocol V6 reference, but the

registered regularity magnitude was not confirmed. A preregistered development screen then tested three lower-actor freeze schedules; none restored the registered long-training effect, so confirmation was not authorized.

The external comparison clarifies the practical meaning of these results. FreqDuet was more regular than fixed headway and improved passenger journey over the weaker rule-holding and rule-MPC baselines, yet fixed headway remained better for passenger journey, holding burden, and fleet readiness. The study therefore supports a registered gate-positive short-horizon result for the complete selected controller. Because the Holm-adjusted sign-flip result was $p = 0.125$, it does not establish a familywise-significant effect, long-run robustness, or universal superiority. A fresh factorial confirmation and matched route-day evaluation are required to isolate frequency allocation. A passenger-centered objective must also determine whether improved regularity can be converted into a passenger-service benefit.

7. Data and Code Availability

The repository contains the compact CSV and JSON artifacts supporting the V8 confirmation, V9 robustness test, V33 timescale screen, and V9 external-baseline comparison. It also records exact seed contracts, resolved configurations, source commits, derived figure data, and assembly scripts. Checkpoints and full training logs are not part of the manuscript bundle. A persistent archival identifier will be added after the manuscript scope and release snapshot are frozen.

The external realism audit uses bounded subsets derived from public MTA AFC and Halifax Transit APC sources. Source endpoints, selection rules, coverage, and derived hourly profiles are recorded in the packaged data manifests. MTA Bus Time route, stop, and AVL records are stored in a separate offline cache; the API credential used for download is neither stored in the repository nor required to rebuild the manuscript from the derived data. These public data support only the descriptive demand-shape audit. They are not same-day AFC/APC/AVL calibration or observed field outcomes for the simulated network.

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